CampusConnect – AI/ML Enhancement

Mini Project Enhancement: AI-powered personalized campus content recommendations

1. Proposed AI/ML Capability

Selected capability: A personalized recommendation component that suggests relevant campus communities, events, posts, and collaboration opportunities to students based on their interests, skills, activity, and interaction history.

This is suitable for CampusConnect because the platform contains multiple types of campus content. As the amount of content grows, students may find it difficult to discover the communities, events, and opportunities most relevant to them. The AI/ML component improves content discovery without changing the existing core communication and networking features.

2. Identification of the Problem Suitable for AI/ML

- Students may have difficulty finding relevant communities, events, posts, and collaboration opportunities among many available items.
- A purely rule-based feed may show the same or broadly popular content to many users and may not reflect individual interests.
- Manual filtering requires users to repeatedly search for skills, topics, clubs, or events.
- A recommendation system can learn patterns from user interests and interactions and produce a personalized list.

3. AI/ML Technique and Justification

Technique: Content-based recommendation with a similarity model, optionally combined with simple interaction signals. A practical implementation could use TF-IDF vectors and cosine similarity or sentence embeddings from a pre-trained model.

Technique	Purpose in CampusConnect
TF-IDF / text embeddings	Convert community, event, and post descriptions into numerical representations.
Cosine similarity	Measure how closely an item matches a student's interests or previously engaged content.
Optional interaction weighting	Increase relevance using actions such as viewed, liked, saved, followed, or attended.

Justification: This approach is appropriate for a mini project because it can use existing CampusConnect data, does not require a very large labelled dataset, is relatively explainable, and can be integrated as an additional service without replacing the existing application architecture.

4. Integration with the Existing CampusConnect Application

The proposed AI component would be added as an independent recommendation service around the existing CampusConnect application. The current application can continue handling authentication, profiles, communities, events, posts, connections, notifications, and messaging.

- Frontend: A new “Recommended for You” section can be displayed on the home/discover page.
- Backend: FastAPI can expose an endpoint such as /recommendations for the logged-in user.

- Recommendation service: The service receives the user's interest/profile information and candidate campus content, calculates relevance, and returns ranked recommendations.
- Database: Existing MySQL data such as user skills, community descriptions, event descriptions, posts, and selected interaction records can provide the inputs.
- Authentication: The existing JWT-based authentication can identify which user's recommendations should be generated.

Illustrative flow

Student → React frontend → FastAPI recommendation endpoint → recommendation service/model → MySQL/content data → ranked results → React “Recommended for You” section.

5. Input and Output Handling

Possible inputs:

- Student profile information and selected skills.
- Interests or topics represented in the student's profile.
- Descriptions/titles of available communities, events, posts, or collaboration opportunities.
- Optional interaction history such as viewed, liked, saved, followed, or attended items.

Expected output:

A ranked list of relevant campus items, for example: Community A (0.91 relevance), Event B (0.84), Post C (0.78), and Collaboration D (0.74). The numerical similarity score would be used internally; the interface can simply display the recommended items and a short reason such as “Matches your interest in Web Development.”

6. Testing with Suitable Examples / Dataset

For demonstration, a small structured dataset can be prepared from sample CampusConnect records. No production-scale dataset is required for this academic enhancement.

Student profile	Available item	Expected behaviour
Java, backend, APIs	Backend Development Community	High relevance
UI/UX, React, frontend	Frontend Design Workshop	High relevance
Machine Learning, Python	ML Study Group	High relevance
Photography, media	Photography Club Event	High relevance
Java, backend	Unrelated Sports Event	Low relevance unless other signals support it

7. Evaluation of the AI/ML Component

The recommendation component can be evaluated using both offline metrics and functional testing. For a mini project, the following measures are sufficient:

- Precision@K: proportion of the top-K recommendations considered relevant.
- Recall@K: proportion of relevant items that appear in the top-K results.
- Click/engagement rate: percentage of recommendations that users open, save, follow, or otherwise interact with.

- Relevance testing: manually verify whether recommendations match the student's stated skills/interests.
- Response time: check that recommendations can be generated within an acceptable time for normal application use.

8. Limitations and Possible Errors

- Cold-start problem: New users with little profile or interaction data may receive generic recommendations.
- Limited data: A small dataset may not capture the full variety of student interests.
- Incorrect profile information: Recommendations can be poor if users enter inaccurate or outdated skills/interests.
- Popularity bias: Frequently interacted-with content may receive more exposure than less popular but relevant content.
- Semantic limitations: Simple TF-IDF may miss deeper meaning or synonyms; embeddings can improve this but add model and computation requirements.
- Feedback loop: Recommending only similar content can reduce discovery of new topics and communities.

9. Privacy, Security, and Ethical Considerations

- Use only the minimum user data required for recommendations.
- Do not expose one student's private messages, private profile information, or personal interaction history to another student.
- Keep recommendation inputs protected using the application's existing authentication and authorization mechanisms.
- Do not use sensitive personal attributes for recommendation decisions unless there is a clearly justified and ethically appropriate requirement.
- Inform users that recommendations are automated and may contain errors.
- Provide a way to control or reset recommendation-related personalization where practical.
- Avoid using recommendation logic to unfairly suppress particular communities, users, or legitimate campus content.

10. Demonstration of the Complete Integrated Application

The final demonstration would show CampusConnect operating normally with the new AI/ML feature added. The demonstration can follow this sequence:

1. Log in as a student using the existing authentication system.
2. Open the home/discover page and show the new “Recommended for You” section.
3. Show that the recommendations correspond to the student's skills/interests.
4. Change the student's profile interests or use a second sample student account and show that the recommendation list changes.
5. Open a recommended community/event/post and demonstrate that the existing CampusConnect functionality continues to work.
6. Briefly explain the model input, recommendation process, output, evaluation method, and limitations.

11. Proposed Academic Outcome

The enhancement demonstrates how an existing full-stack campus application can be extended with a practical AI/ML capability rather than adding AI only for demonstration purposes. The recommendation component addresses the real problem of information discovery, uses an appropriate and explainable ML approach, integrates with the existing CampusConnect architecture, accepts application data as input, produces personalized recommendations, and can be evaluated using relevance and engagement measures.

Note: This document describes the proposed AI/ML enhancement and its implementation plan only. No actual AI/ML implementation, model training, API integration, or code is required for this submission.